CellPress

Trends in
Ecology & Evolution

Opinion

Resilience and robustness: from sub-organismal responses to communities

Maren N. Vitousek [1,2,*], Conor C. Taff [1,2], and Jessie L. Williamson [1,2]

Coping with challenges is essential to life on earth. Determining the processes that generate resilience and robustness to disturbance across levels of biological organization is increasingly important as the pace of global change accelerates; however, to date, multiscale models have primarily focused on population to ecosystem scales. In this opinion article we combine conceptual models from different fields to develop a unified a framework of resilience and robustness that explicitly links sub-organismal responses with higher-level outcomes. This framework predicts that interactions among sub-organismal response components – including their temporal dynamics and the plasticity of homeostatic regulatory networks – are key drivers of current and future resilience.

Highlights

Resilience and robustness can be unifying principles in biology, and are particularly important in a rapidly changing world.

Multiscale models of resilience in ecology and evolution have primarily focused on higher levels of organization (populations to ecosystems).

Sub-organismal responses and the properties of the homeostatic regulatory networks that connect them differ from the mechanisms that underlie networks at other levels of organization; thus, findings at one level cannot necessarily be extrapolated to other levels.

Connecting models of dynamic sub-organismal responses with network models of organismal function will enable a deeper understanding of individual variation in resilience and robustness, and its population and community level effects.

Resilience and robustness in sub-organismal processes: why do they matter?

The ability to cope with challenges is fundamental to life on earth; consequently, **resilience** (see Glossary) is a central concept across levels of biological organization and fields of study. In a rapidly changing world, it is also increasingly important to understand what drives differences in the **susceptibility** of individuals, populations, communities, and ecosystems to disturbance [1,2]. The broad importance of resilience and **robustness** across living systems and their potential to be unifying concepts in biology has spurred an explosion of research into their predictors, and calls for deeper understanding of how these concepts scale across levels of biological organization (e.g., [3–5]).

Despite this interest, multiscale models of resilience and robustness in ecology and evolution have focused largely on higher levels of organization (i.e., population to ecosystem scales [6,7]) – from Holling's classic ball-and-cup analogy [8], which conceptualized resilience as the capacity of an ecological system to return to its original state after a perturbation – to subsequent conceptual and empirical models [9–14]. Recently, a promising direction has been the use of network-based approaches to model how the structure and function of ecological networks affects responses to disturbance [15,16]. Incorporating tools and approaches from a diversity of fields, including engineering, macroecology, and systems biology, these models have connected processes occurring at one level of biological organization with resilience and robustness at the level above.

A broad generalization of both conceptual and empirical models is that responses at lower levels can alter higher-level outcomes [3]. Resilience or robustness at lower levels often predicts resilience or robustness at the level above; susceptibility at lower levels decreases the likelihood of resilience or robustness at higher levels, but susceptibility in individual **nodes** in a network can also be buffered by network properties (e.g., **redundancy**, **connectivity**) [3,4]. Identifying the processes that generate resilience at lower levels is therefore crucial for predicting organismal resilience and its capacity to evolve, and likely also has implications for how communities and

[1]Department of Ecology and Evolutionary Biology, Cornell University, Ithaca, NY 14853, USA
[2]Cornell Laboratory of Ornithology, Ithaca, NY 14850, USA

*Correspondence:
mnv6@cornell.edu (M.N. Vitousek).

© 2025 Published by Elsevier Ltd.

Check for updates

ecosystems respond to disturbance (Figure 1). However, we currently lack a framework that explicitly defines and predicts resilience and robustness across sub-organismal responses (e.g., cellular, hormonal, transcriptomic) and generates testable predictions about their effects on resilience and robustness at higher levels.

Because the mechanisms that underlie networks differ across scales in meaningful ways, we cannot simply extrapolate findings from one level of biological organization to another. For example, below the level of organisms, network nodes (sub-organismal processes) are linked by homeostatic mechanisms that function in coordinated ways to preserve the stability of the network [17,18]; this lacks a parallel at higher levels. Similarly, natural selection operates at the organismal level, although it has implications for levels above and below (Figure 1). Because of this, the structure and properties of sub-organismal networks can evolve jointly through natural selection (e.g., favoring the redundancy in metabolic networks that promotes robustness [3,19]). By contrast, natural selection does not operate directly on the integrated functions of community- or ecosystem-level networks. Connecting realistic models of sub-organismal processes and the networks that connect them with their higher-level outcomes can identify drivers of individual variation in resilience, link sub-organismal responses with the resilience and robustness of populations and communities, and help to illuminate how resilience and robustness scale across levels of biological organization.

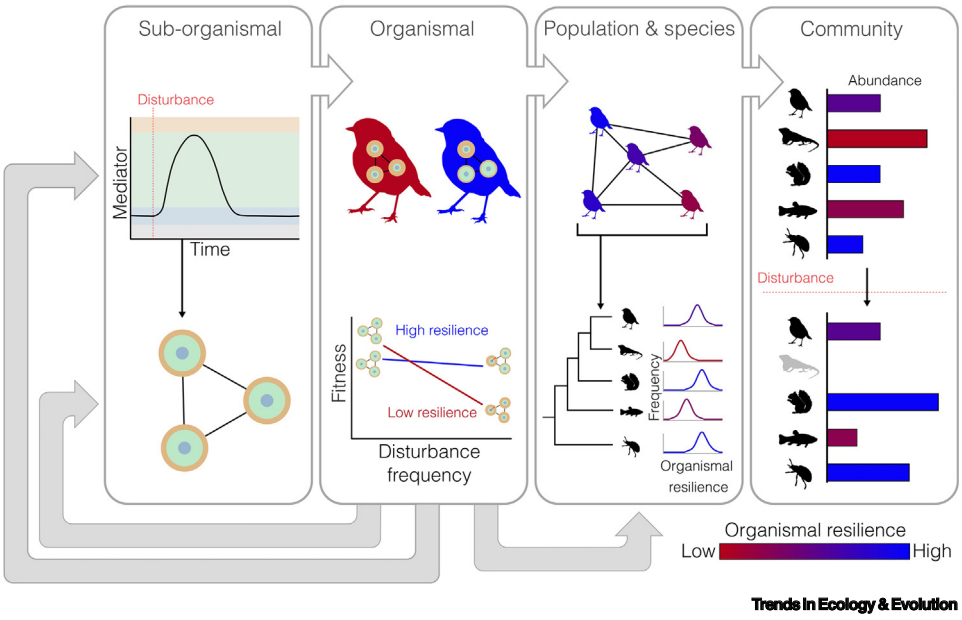


Figure 1. Resilience at sub-organismal and organismal levels affects resilience across scales of biological organization. At the sub-organismal level, resilience is influenced by the temporal dynamics of physiological responses to disturbance and their reactive scopes, and by the structure and properties of the homeostatic regulatory network that connects response systems (network nodes). Variation in the capacity of the homeostatic regulatory network to maintain essential functions through disturbance generates variation in resilience at the organismal level; individual variation in resilience can be maintained by trade-offs across disturbance regimes (e.g., variation in the frequency, severity, or nature of stressors). Resilience to a specific disturbance regime at the population or species level (e.g., changes in population size) is affected by the frequency distribution of resilience phenotypes within the population, interactions among individuals in the population network, and evolutionary history. This in turn can affect the composition of ecological communities [55], and their capacity to maintain essential ecosystem functions following a disturbance, by changing the presence or abundance of species. Selection operating at the level of individuals shapes sub-organismal response dynamics, the structure and properties of homeostatic regulatory networks, and population- or species-level resilience. Silhouette images are from www.phylopic.org.

Glossary

Connectivity: the number and strength of connections among nodes in homeostatic regulatory networks.
Homeostatic overload: the state triggered by levels of a physiological mediator or rates of a process that exceed the reactive scope, perturbing the network and causing phenotypic damage.
Homeostatic regulatory network: the network of sub-organismal processes that maintains homeostasis, a state of healthy physiological equilibrium.
Nodes: components of a network that are connected to other components. Sub-organismal processes involved in maintaining homeostasis are nodes within the homeostatic regulatory network. Individuals are nodes within a population network.
Predictive homeostasis: the fluctuating range of rates of a physiological process or levels of a mediator that constitute normal 'baseline' function.
Reactive homeostasis: levels or rates of physiological process elicited by a disturbance or environmental change that exceed the range of predictive homeostasis but remain below the threshold of homeostatic overload; it induces physiological, behavioral, and transcriptomic responses that support the maintenance of or return to homeostasis.
Reactive scope: the flexible range of levels or rates of a physiological process or its mediators that characterize normal, non-pathological function; includes the ranges of predictive and reactive homeostasis.
Redundancy: a network property characterized by similar or overlapping functions in nodes. More redundant networks have more nodes with similar functions.
Resilience: returning to a pre-disturbance state following a disturbance. At the sub-organismal level, resilient responses temporarily exceed but ultimately return to the range of predictive homeostasis following a disturbance.
Response scope: the magnitude of within-individual changes in the levels of a mediator or rates of a process in response to a disturbance or across an environmental gradient.
Robustness: the maintenance of a consistent state following a disturbance.

Modeling and interpreting resilience and robustness

Organismal resilience and robustness depend on the function of numerous interacting network components. Predicting these outcomes requires a realistic model of responses within key sub-organismal nodes (e.g., physiological integrators [18–20]) and of the connections among them.

Conceptual models developed in the field of stress physiology provide a framework for characterizing sub-organismal responses and their effects on network function. In the **reactive scope** model [21], which extends earlier models of allostasis (e.g., [22]), mediators of the physiological stress response, or rates of a process, fluctuate daily or seasonally within a 'baseline' range (**predictive homeostasis**) (Box 1). Following an unpredictable or threatening challenge, mediator levels enter the range of **reactive homeostasis**, triggering phenotypic and sometimes transcriptomic responses that facilitate the return to homeostasis. This kind of response perturbs the homeostatic network, but the network is resilient to the disturbance. If the challenge persists and levels of the mediator exceed the reactive scope, **homeostatic overload** occurs; in this range, mediator levels cause pathological damage [21], impairing the function of other nodes. Individual nodes operating in homeostatic overload could trigger connected nodes to enter homeostatic overload or to decrease their reactive scope (Box 1). If the response persists and the network is not robust to this disturbance (e.g., through redundancy or changes in network connectivity), the **homeostatic regulatory network** will ultimately fail.

A key feature of this model is that physiological responses with identical **response scopes** (e.g., changes in mediator levels) can have different effects depending on the breadth of the reactive scope. Responses that remain within the reactive scope trigger phenotypic changes aimed at maintaining or returning to homeostasis; responses that exceed the reactive scope impose a multitude of costs on other physiological systems, threatening the stability of that response system [21]. In other words, the reactive scope of a sub-organismal response impacts how it influences other components of homeostatic networks. Organismal resilience or robustness is ultimately determined by a combination of response dynamics in individual nodes, the breadth of the reactive scope, and network properties. To date, this model has been applied predominantly to the regulation of glucocorticoids, neuroendocrine mediators of the physiological stress response secreted by the hypothalamic–pituitary–adrenal (HPA) axis, but it is similarly relevant to other physiological mediators and processes [21]. Because maintaining resilience likely depends on the activity of mediators within the primary stress response systems (e.g., HPA axis, catecholamine-mediated responses, immune system, and central nervous system), the dynamics of responses within these core systems, their associated costs, and their effects on other physiological mediators involved in the maintenance of homeostasis, represent particularly promising targets.

Sub-organismal processes can be characterized as robust or resilient depending on the outcome of their response to disturbance. However, the same outcomes can be generated by different response patterns. Research in the field of stress physiology indicates that response dynamics influence not only the impact of the current disturbance but also the likelihood of future resilience (Box 1). A common response to repeated or persistent disturbance is a reduction in the scope of the physiological response to that challenge [23]. This reduction can result from habituation to the disturbance, such that it is no longer perceived as a challenge, but it can also be a mechanism to compensate for a shrinking reactive scope and minimize the damage of an extended stress response. In the latter case, individuals that appear to exhibit a more resilient response (with levels that return closer or more quickly to a baseline state) could paradoxically experience greater constriction of their reactive scope. Critically, the reduction in response scope during this type of acclimation does not extend across stressor types [23]. Thus, individuals that reduce their response

At the sub-organismal level, a robust response to disturbance is characterized by the maintenance of a process or its physiological mediators within the range of predictive homeostasis.
Stressor priming: a recalibration of the stress response system triggered by mounting a physiological stress response.
Susceptibility: exhibition of an altered state following a disturbance; a response pattern that is neither robust nor resilient.

to one kind of challenge could have a higher likelihood of homeostatic overload and lower resilience to other challenges. For example, trout facing persistent heavy metal contamination reduce their hormonal response to contamination, but if faced with an additional stressor, they mount a strong hormonal response followed by homeostatic failure and death [24,25]. As this scenario

Box 1. Predicting the likelihood of resilience using a multidimensional framework of response dynamics

Conceptual models of the physiological stress response can be combined with network models of organismal function to provide a framework through which the dynamics of sub-organismal responses (e.g., neuroendocrine, metabolic, cellular), including their resilience or robustness, can be explicitly linked with resilience or robustness at higher levels. In the reactive scope model [21], responses within the reactive homeostasis range cause a narrowing of the reactive scope, lowering the threshold of homeostatic overload within a node (Figure I). This results from a combination of direct costs of the mediator itself and the 'wear and tear' induced by compensatory adjustments in other network components. Homeostatic overload in one node increases the likelihood of homeostatic overload at connected nodes; it could also induce changes in network structure or properties (Figure II). Through the narrowing of the reactive scope within network nodes, even resilient responses to challenges can decrease the likelihood of future resilience at both sub-organismal and organismal levels (Figure II). When a perturbation ends, the reactive scope begins to recover. Thus, depending on the frequency and severity of challenges, resilience can be maintained despite persistent stressors.

The likelihood that a given sub-organismal response system will exhibit resilience to a disturbance is affected by whether and how much the response exceeds the reactive scope, and by the capacity of the homeostatic regulatory network to buffer dysregulation within a node and avoid homeostatic failure. Whether a response exceeds the reactive scope depends on the scope of the response and its temporal dynamics (e.g., the speed of activation and recovery rate), as well as the breadth of the reactive scope and how it changes over time (e.g., the slope of decrease in response to stressors and of increase during recovery) (Figure III). Variation among individuals, populations, and species in the scope of sub-organismal responses is commonly measured [55–58], but the temporal dynamics of physiological responses have received much less attention [47,48]. Furthermore, while these measures interactively affect the magnitude of departure from the reactive scope, individual components of physiological responses are typically measured in isolation, which has likely contributed to prior findings of conflicting relationships with proxies of organismal resilience [59,60]. Particularly important future directions include elucidating the interactions among response dynamics and linking them with resilience in sub-organismal response systems, and with organismal resilience and robustness. Mathematical models are likely to be valuable companions to empirical data in this endeavor [61,62].

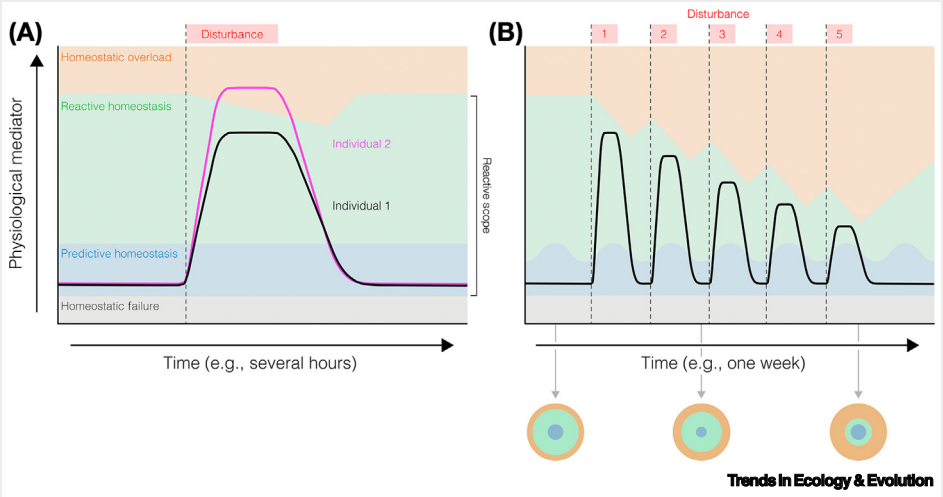


Figure I. (A) In the reactive scope model, physiological stress responses in the range of reactive homeostasis or homeostatic overload trigger a constriction of the reactive scope. (B) Repeated disturbances experienced by the same individual trigger a constriction in the reactive scope (blue and green regions), which is sometimes accompanied by a compensatory reduction in the scope of the response to that type of disturbance (the magnitude of the change in mediator levels). The breadth of the predictive homeostasis range may change seasonally or daily, as shown here. Circles below the x-axis illustrate the breadth of the reactive scope at each point in time; this graphical representation is suitable for inclusion in network diagrams.

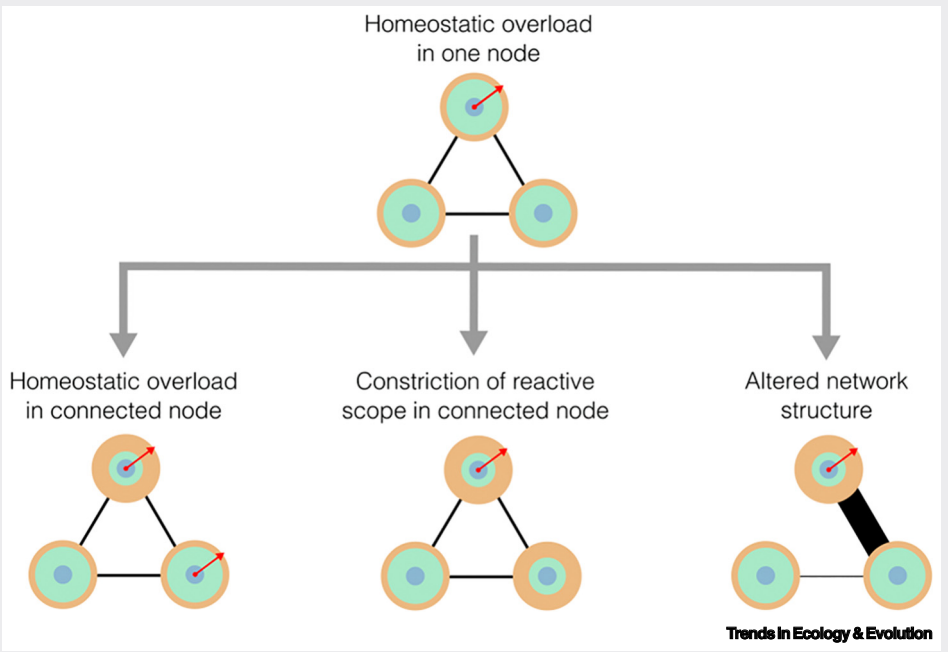


Figure II. Possible non-mutually exclusive effects of homeostatic overload in one node on the function of other nodes and network structure. Circles represent the current state of a node in the homeostatic network. Colors correspond to ranges of the reactive scope; their thickness indicates the breadth of the reactive scope (blue and green regions). Red arrows illustrate mediator levels in homeostatic overload. The width of black lines connecting circles represents the strength of network connections. Potential outcomes illustrated include connected nodes entering homeostatic overload, connected nodes constricting their reactive scope (e.g., from responses in the reactive homeostasis range), or a change in network structure.

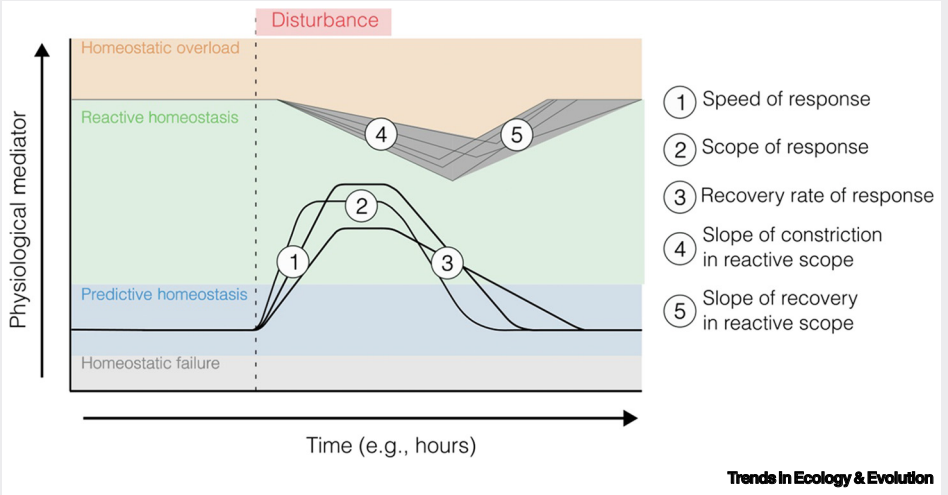


Figure III. Whether a response exceeds the reactive scope depends on multidimensional response dynamics. These phenotypically flexible response dynamics can vary within and among individuals (e.g., based on genetic background, developmental environment, and prior exposure), and across populations and species.

illustrates, the resilience or robustness of sub-organismal responses should not be interpreted in isolation, but within a multidimensional framework of response dynamics that incorporates responsiveness (e.g., the scope and speed of responses to a perturbation) and the reactive scope.

Most research on the mechanisms of organismal resilience has focused on inter-individual variation in specific components of sub-organismal responses [26–28]. Few studies have characterized multidimensional response dynamics, assessed sub-organismal resilience or robustness, or tested their effects on network function and organismal susceptibility. Similarly, despite the impact of the reactive scope model [21] on the field of stress physiology, empirical tests of predictions about how reactive scopes change over time are lacking. Because the reactive scope modifies the phenotypic consequences of a response, changes in its breadth could be inferred by assessing how responses of different levels affect other network hubs and alter the structure and properties of the homeostatic regulatory network. The physiological and behavioral regulatory networks that maintain homeostasis and support performance have long been of interest to organismal biologists [17,18,20]; however, significant gaps remain in defining their properties, and in testing how sub-organismal response dynamics influence network-level resilience and robustness, and conversely, how network properties can affect response dynamics (including reactive scopes) within individual nodes. This could be addressed by measurements or manipulation of response dynamics within key nodes coupled with coordinated and repeated measurements of key traits on the timeline of interest, including network hubs (e.g., through physiological biologging) [29] or measurements of gene or protein expression associated with homeostatic regulation (Box 2). Testing this integrated framework

Box 2. The evolution of resilience and robustness

The evolvability of resilience and robustness are poorly understood, although studies at higher levels of organization often assume that these response patterns can evolve. The selective benefit of exhibiting a particular response pattern should differ by environment, depending on the temporal dynamics, predictability, and severity of stressors, and stressor type. Although resilience and robustness are often considered to be part of a continuum of resistance to disturbance, these response patterns can be underlain by different mechanisms with different selective pressures [5,63]. They are likely constrained by trade-offs, including between resilience and robustness, resilience and maximal performance, and robustness and recovery rate [5,12]. Network structure is also likely shaped by trade-offs, including between the number and connectivity of nodes and the local stability of networks [5,64].

Although resilience and robustness are widely considered to be desirable outcomes, sub-organismal resilience will not always be favored by selection, even in the absence of trade-offs [1]. For example, when disturbances are frequent, it may be adaptive for a response to transition to an alternative state, or for the properties of the homeostatic network to shift, rather than to exhibit resilience. The evolvability of resilience and robustness will likely differ based on their genetic architecture [65,66]; failing to consider genetic mechanisms could lead to erroneous conclusions about how selection acts on resilience and robustness.

Because the breadth of the reactive scope affects how a given response influences the network, and organismal resilience (Box 1), determining how genes and environment interactively influence reactive scopes – and how this varies within and among populations across timescales – are important future directions. Directly assessing the reactive scope by coupling experimental manipulation of the response (e.g., altering mediator levels to induce homeostatic overload) with repeated measures of network-level effects (e.g., using -omics technologies) is a promising approach. Within systems, network-level analyses could identify connected nodes and test whether their response dynamics change in response to homeostatic overload at a given node [67]. Homeostatic overload involves the accrual of damage caused by the mediators of the physiological stress response, when normal defense mechanisms are no longer sufficient to counteract their costs. Existing knowledge of the phenotypic costs of specific mediators in well-studied systems could be used to identify likely indicators (e.g., homeostatic overload in the HPA axis can trigger muscle breakdown, neuronal atrophy, insulin resistance, and immunosuppression); network-level analyses could be leveraged to identify additional targets for investigation. While some of these approaches will be challenging to implement within individuals, comparisons among clonal lines or strains, and among populations or species that inhabit different environments, could be particularly fruitful. Identifying trade-offs associated with response dynamics can also help to illuminate trade-offs in organismal and population-level resilience. For example, do trade-offs between robustness and the rate of recovery of the reactive scope generate trade-offs in the capacity to cope with disturbances at different temporal scales?

through targeted data collection could illuminate the capacity of homeostatic regulatory networks to retain the essential functions that support organismal performance in the face of disturbance, generating the demographic patterns that drive population-level resilience or robustness.

Bridging fields to illuminate the mechanisms of organismal resilience

Although resilience and robustness are relevant across biological disciplines, what constitutes a 'resilient organism' differs by field [3,9]. For example, in evolutionary terms, organismal resilience and robustness are defined by changes in fitness [30,31]; physiological studies tend to focus on homeostatic maintenance [32], and biomedical research often defines resilience with respect to neuropsychiatric disorders, including the absence of post-traumatic stress disorder (PTSD) after trauma [33,34]. Regardless of the specific measure used, organismal resilience is likely an emergent property influenced by the function of many sub-organismal processes.

Key components of the physiological and transcriptomic response to many stressors are challenge-specific (e.g., the increase in hemoglobin concentration in response to hypoxia [35,36]). Such responses are likely regulated by trade-offs with their immediate costs, and with susceptibility to opposing challenges (e.g., trade-offs in the capacity to cope with heat and cold [37]). At the same time, both biomedical and eco-evolutionary studies suggest that some phenotypes are more resilient or robust than others across a larger suite of challenges [38,39]. This could result from differences in the perception of stressors, response dynamics within key nodes of the network, or the properties of the network that buffer it from localized disturbance [40–43].

Evidence across fields suggests that the speed of onset and recovery rate of sub-organismal responses can predict organismal susceptibility. For example, in some reptiles and birds, more rapid resilience in the glucocorticoid response (i.e., stronger negative feedback following a standardized acute stressor) positively predicts resilience or robustness to a persistent disturbance at the organismal level (measured as the ability to survive or continue reproducing in natural populations) [27,31,44]. Similarly, in humans and rodent models of neuropsychiatric disorders, variation in HPA axis negative feedback is associated with behavioral resilience (e.g., in depression, anxiety, and PTSD) [34,45]. Reductions in the initial speed of physiological responses to disturbance, as well as their recovery rate, have also been linked with negative health outcomes in humans [29], and across species of birds, faster hormonal responses appear to be favored in more extreme thermal environments [46].

Despite these patterns, the temporal dynamics of responses are especially understudied at the sub-organismal and organismal levels. Careful consideration of measurement timing is necessary for differentiating between resilience and susceptibility and between resilience and robustness, which can result from opposing mechanisms and may be regulated by trade-offs (Box 2). Yet these outcomes are commonly assessed using a single post-disturbance measure, or a sampling scheme that is not well suited for differentiating these response patterns [33,47]. Even among the best-studied sub-organismal responses, like the glucocorticoid response, significant gaps remain in characterizing their temporal dynamics and linking those dynamics to resilience and robustness at higher levels [47,48].

Several response systems linked with resilience in biomedical research have received much less attention in eco-evolutionary studies of resilience, including immune regulation, neuroinflammatory processes, and the microbiota–gut–brain axis [33,34,49]. Testing whether the dynamics of

associated sub-organismal responses, including their resilience and robustness, predict organismal resilience in natural contexts will likely be illuminating. Another important future direction is testing how various sub-organismal response systems work together to alter organismal resilience and robustness to specific challenges, especially those that are increasingly prevalent in a changing world (e.g., thermal challenges [50,51]).

Connecting plasticity with resilience and robustness

Phenotypic plasticity and resilience are linked because resilient responses are characterized by temporary changes in phenotype. While plasticity – including reversible flexibility – within sub-organismal nodes has been widely studied, our understanding of plasticity in the structure and properties of homeostatic regulatory networks is comparatively thin. Nevertheless, it is clear that network connections can change over time. For example, while prolonged activation of the HPA axis (i.e., homeostatic overload) often impedes the secretion of gonadal hormones – thereby impairing or preventing reproduction – the presence and strength of this effect varies across social and environmental contexts (e.g., [52]); thus, the structure of the neuroendocrine network that underlies robustness to stress is plastic. Similarly, metabolomic approaches have begun to elucidate how the structure of metabolic networks changes during acclimation to stressors (e.g., [53]).

There is also substantial evidence of plasticity in resilience and robustness at the organismal level. A key priority is connecting plasticity in sub-organismal processes with organism-level outcomes (Box 3) to determine when plasticity at lower levels promotes resilience or robustness at higher levels. Broadly, we predict that sub-organismal flexibility will increase organismal resilience or robustness and be favored by selection (if it has a heritable basis) when flexibility is sufficiently fast relative to the timescale on which environmental changes occur, and when the costs or limits to sub-organismal flexibility are relatively low [48,54]. In heterogeneous environments, developmental plasticity or other lasting changes in sub-organismal responses are likely to be favored when these conditions are met, and when cues reliably indicate subsequent environmental conditions.

Box 3. Plasticity and the priming of resilience

Resilience and robustness at the organismal and sub-organismal levels can be shaped by environment and prior experience. Experiencing and responding to stressors often increases organismal susceptibility to future challenges [22,68] (Box 1). But exposure to challenges during development [69–71] or in adulthood [72,73] can also increase future resilience or robustness. Identifying the sub-organismal changes that underlie adaptive plasticity is important for understanding the drivers of resilience and robustness. Long-term changes in phenotype triggered by experience in adulthood are relatively poorly understood but can reflect the lasting effects of a targeted response to a specific stimulus (e.g., changes in enzymatic activity following thermal stress [73]) or result from individuals learning about the nature of a specific threat and altering their response accordingly [74]. However, experiments in a variety of species – from those with relatively advanced cognitive abilities to those that lack nervous systems – suggest that exposure to one type of disturbance can sometimes reduce susceptibility to other types of challenge [38,72]. This generalized effect could result from **stressor priming** [40,75], a recalibration of the stress response system triggered by mounting a physiological stress response that results in more efficient or appropriate responses to future challenges, and potentially an increase in resilience or robustness [76]. For example, *Arabidopsis thaliana* primed with β-aminobutyric acid (an amino acid induced by stressor exposure) increase the speed of the response to a subsequent drop in humidity: stomata are closed faster, and the plants are more resilient to drought [77]. This could occur through changes in response dynamics within nodes, particularly those that regulate and integrate phenotypic responses to stress. The generalized effects of stressor priming suggest that it may also involve lasting changes in the structure or properties of networks [75]. Testing when and how exposure to environmentally relevant challenges in adulthood induces short-term flexibility or lasting changes in the resilience or robustness of sub-organismal responses, or in the interactions among homeostatic regulatory network nodes, are important future directions for research. Promising candidate mechanisms include epigenetic processes and the regulation of mediators of the generalized physiological stress response [39,76,78,79].

Concluding remarks

A unified multiscale framework of resilience and robustness requires an understanding of the processes that generate these outcomes within each level of biological organization, and how they affect resilience and robustness at the levels above. Connecting conceptual models of the function of sub-organismal mediators of the response to disturbance with their effects on the homeostatic regulatory network provides testable predictions about the drivers of resilience and robustness at sub-organismal and organismal levels. In this framework, the likelihood of resilience is affected by whether and how much sub-organismal responses exceed their reactive scope, and by the capacity of the homeostatic regulatory network to buffer dysregulation within nodes in homeostatic overload.

Key priorities for future research (see Outstanding questions) include testing how interacting response dynamics, including the speed and scope of responses and their reactive scopes, influence resilience in sub-organismal responses, especially those that function as hubs in the homeostatic regulatory network. Characterizing reactive scopes – an important but understudied component of sub-organismal responses – and their temporal dynamics (e.g., rates of constriction in response to disturbance and expansion during recovery) will be valuable for predicting resilience and robustness to disturbance. Evaluating how sub-organismal response dynamics affect the structure and properties of homeostatic networks, and ultimately demographic rates, and connecting plasticity in sub-organismal nodes and networks with higher-level resilience and robustness are important future directions, particularly in the face of rapid global change. Determining the genetic architecture and fitness effects of sub-organismal response dynamics and of the properties of the networks that connect them, will facilitate a deeper understanding of the evolution of resilience and robustness, and how they scale across levels of biological organization.

Outstanding questions

How do interactions among response dynamics influence current and future resilience?

What mechanisms and trade-offs underlie variation in the temporal dynamics of reactive scopes?

When and how do network properties change when responses exceed the reactive scope? Do the structure and properties of networks influence response dynamics (including the reactive scope) within nodes, altering the likelihood of homeostatic overload?

What are the mechanisms of generalized resilience (resilience to multiple stressors)?

Does stressor priming influence future resilience or robustness?

How do the structure and properties of homeostatic regulatory networks, and their plasticity, influence organismal resilience and robustness?

How do populations of widely distributed species vary in their daily and seasonal responses to different environments?

How does the strength of selection on response dynamics, including robustness and resilience, differ across space and time?

How does resilience evolve? What is its genetic architecture? How is it affected by trade-offs? When is it not favored by selection?

Acknowledgments

M.N.V. and C.C.T. were supported by NSF-IOS 2028337. J.L.W. was supported by NSF-DBI-2208924 and a Cornell Laboratory of Ornithology Edward W. Rose Postdoctoral Fellowship. We are grateful for suggestions from two reviewers that improved the manuscript.

Declaration of interests

The authors declare no competing interests.

References

1. Cinner, J.E. and Barnes, M.L. (2019) Social dimensions of resilience in social-ecological systems. *One Earth* 1, 51–56
2. Grafton, R.Q. *et al.* (2019) Realizing resilience for decision-making. *Nat. Sustain.* 2, 907–913
3. Crespi, E. *et al.* (2022) Resolving the rules of robustness and resilience in biology across scales. *Integr. Comp. Biol.* 61, 2163–2179
4. Davis, J.E. *et al.* (2021) Toward a universal theoretical framework to understand robustness and resilience: from cells to systems. *Front. Ecol. Evol.* 8, 579098
5. Inouye, B.D. *et al.* (2022) Trade-offs among resilience, robustness, stability, and performance and how we might study them. *Integr. Comp. Biol.* 61, 2180–2189
6. Capdevila, P. *et al.* (2020) Towards a comparative framework of demographic resilience. *Trends Ecol. Evol.* 35, 776–786
7. Thorogood, R. *et al.* (2023) Understanding and applying biological resilience, from genes to ecosystems. *Npj Biodivers.* 2, 16
8. Holling, C.S. (1973) Resilience and stability of ecological systems. *Annu. Rev. Ecol. Syst.* 4, 1–23
9. Angeler, D.G. and Allen, C.R. (2016) Quantifying resilience. *J. Appl. Ecol.* 53, 617–624
10. Baho, D.L. *et al.* (2017) A quantitative framework for assessing ecological resilience. *Ecol. Soc.* 22, art17
11. Falk, D.A. *et al.* (2019) Scaling ecological resilience. *Front. Ecol. Evol.* 7, 275
12. Gladstone-Gallagher, R.V. *et al.* (2019) Linking traits across ecological scales determines functional resilience. *Trends Ecol. Evol.* 34, 1080–1091
13. Ingrisch, J. and Bahn, M. (2018) Towards a comparable quantification of resilience. *Trends Ecol. Evol.* 33, 251–259
14. Lamothe, K.A. *et al.* (2019) Linking the ball-and-cup analogy and ordination trajectories to describe ecosystem stability, resistance, and resilience. *Ecosphere* 10, e02629
15. Enquist, B.J. *et al.* (2024) Scaling approaches and macroecology provide a foundation for assessing ecological resilience in the Anthropocene. *Philos. Trans. R. Soc. B Biol. Sci.* 379, 20230010
16. Jeynes-Smith, C. *et al.* (2024) Identifying and explaining resilience in ecological networks. *Ecol. Lett.* 27, e14484
17. Cohen, A.A. *et al.* (2012) Physiological regulatory networks: ecological roles and evolutionary constraints. *Trends Ecol. Evol.* 27, 428–435
18. Martin, L.B. and Cohen, A.A. (2014) Physiological regulatory networks: the orchestra of life? In *Integrative Organismal Biology* (Martin, L.B. *et al.*, eds), pp. 137–152, Wiley
19. Jeong, H. *et al.* (2000) The large-scale organization of metabolic networks. *Nature* 407, 651–654

20. Fuxjager, M.J. *et al.* (2023) Systems biology as a framework to understand the physiological and endocrine bases of behavior and its evolution – from concepts to a case study in birds. *Horm. Behav.* 151, 105340
21. Romero, L.M. *et al.* (2009) The reactive scope model – a new model integrating homeostasis, allostasis, and stress. *Horm. Behav.* 55, 375–389
22. Wingfield, J.C. *et al.* (1998) Ecological bases of hormone – behavior interactions: the 'emergency life history stage'. *Am. Zool.* 38, 191–206
23. Romero, L.M. (2004) Physiological stress in ecology: lessons from biomedical research. *Trends Ecol. Evol.* 19, 249–255
24. Norris, D.O. *et al.* (1999) Impaired adrenocortical response to stress by brown trout, *Salmo trutta*, living in metal-contaminated waters of the Eagle River, Colorado. *Gen. Comp. Endocrinol.* 113, 1–8
25. Cyr, N.E. and Romero, L.M. (2009) Identifying hormonal habituation in field studies of stress. *Gen. Comp. Endocrinol.* 161, 295–303
26. Grueschow, M. *et al.* (2021) Real-world stress resilience is associated with the responsivity of the locus coeruleus. *Nat. Commun.* 12, 2275
27. Vitousek, M.N. *et al.* (2019) Stress resilience and the dynamic regulation of glucocorticoids. *Integr. Comp. Biol.* 59, 251–263
28. Zimmer, C. *et al.* (2020) *FKBP5*: a key mediator of how vertebrates flexibly cope with adversity. *BioScience* 70, 1127–1138
29. Scheffer, M. *et al.* (2018) Quantifying resilience of humans and other animals. *Proc. Natl. Acad. Sci.* 115, 11883–11890
30. Rymer, T.L. *et al.* (2016) Resilience to droughts in mammals: a conceptual framework for estimating vulnerability of a single species. *Q. Rev. Biol.* 91, 133–176
31. Zimmer, C. *et al.* (2019) On again, off again: acute stress response and negative feedback together predict resilience to experimental challenges. *Funct. Ecol.* 33, 619–628
32. Wada, H.H. (2019) Damage-fitness model: the missing piece in integrative stress models. *Stress* 22, 548–562
33. Bhatnagar, S. (2021) Rethinking stress resilience. *Trends Neurosci.* 44, 936–945
34. Kalisch, R. *et al.* (2024) Neurobiology and systems biology of stress resilience. *Physiol. Rev.* 104, 1205–1263
35. Beall, C.M. (2006) Andean, Tibetan, and Ethiopian patterns of adaptation to high-altitude hypoxia. *Integr. Comp. Biol.* 46, 18–24
36. Williamson, J.L. *et al.* (2023) Hummingbird blood traits track oxygen availability across space and time. *Ecol. Lett.* 26, 1223–1236
37. Schou, M.F. *et al.* (2022) Evolutionary trade-offs between heat and cold tolerance limit responses to fluctuating climates. *Sci. Adv.* 8, eabn9580
38. Andrade-Linares, D.R. *et al.* (2016) Microbial stress priming: a meta-analysis. *Environ. Microbiol.* 18, 1277–1288
39. Wu, G. *et al.* (2013) Understanding resilience. *Front. Behav. Neurosci.* 7, 1–15
40. Del Giudice, M. *et al.* (2018) What is stress? A systems perspective. *Integr. Comp. Biol.* 58, 1019–1032
41. Kalisch, R. *et al.* (2019) Deconstructing and reconstructing resilience: a dynamic network approach. *Perspect. Psychol. Sci.* 14, 765–777
42. Word, K.R. *et al.* (2022) Allostasis revisited: a perception, variation, and risk framework. *Front. Ecol. Evol.* 10, 954708
43. Zitnik, M. *et al.* (2019) Evolution of resilience in protein interactomes across the tree of life. *Proc. Natl. Acad. Sci.* 116, 4426–4433
44. Romero, L.M. and Wikelski, M. (2010) Stress physiology as a predictor of survival in Galapagos marine iguanas. *Proc. R. Soc. B Biol. Sci.* 277, 3157–3162
45. Lehrner, A. *et al.* (2016) Cortisol and the hypothalamic–pituitary–adrenal axis in PTSD. In *Posttraumatic Stress Disorder: From Neurobiology to Treatment*, pp. 265, John Wiley
46. Taff, C.C. *et al.* (2022) The relative speed of the glucocorticoid stress response varies independently of scope and is predicted by environmental variability and longevity across birds. *Horm. Behav.* 144, 105226
47. Dupont, L. *et al.* (2024) Beyond reaction norms: the temporal dynamics of phenotypic plasticity. *Trends Ecol. Evol.* 39, 41–51
48. Taff, C.C. and Vitousek, M.N. (2016) Endocrine flexibility: optimizing phenotypes in a dynamic world? *Trends Ecol. Evol.* 31, 476–488
49. Houtz, J.L. *et al.* (2022) The gut microbiome as a mediator of stress resilience: a reactive scope model framework. *Integr. Comp. Biol.* 62, 41–57
50. Crino, O.L. *et al.* (2024) A conceptual framework for understanding stress-induced physiological and transgenerational effects on population responses to climate change. *Evol. Lett.* 8, 161–171
51. Taff, C.C. *et al.* (2024) Endocrine flexibility can facilitate or constrain the ability to cope with global change. *Philos. Trans. R. Soc. B Biol. Sci.* 379, 20220502
52. Wingfield, J.C. and Sapolsky, R.M. (2003) Reproduction and resistance to stress: when and how. *J. Neuroendocrinol.* 15, 711–724
53. Serrano, N. *et al.* (2019) Thermopriming reprograms metabolic homeostasis to confer heat tolerance. *Sci. Rep.* 9, 181
54. Piersma, T. and Drent, J. (2003) Phenotypic flexibility and the evolution of organismal design. *Trends Ecol. Evol.* 18, 228–233
55. Khaliq, I. *et al.* (2024) Warming underpins community turnover in temperate freshwater and terrestrial communities. *Nat. Commun.* 15, 1921
56. Chen, B. *et al.* (2018) Evolution of heat-shock protein expression underlying adaptive responses to environmental stress. *Mol. Ecol.* 27, 3040–3054
57. Hahn, P.G. *et al.* (2019) Population variation, environmental gradients, and the evolutionary ecology of plant defense against herbivory. *Am. Nat.* 193, 20–34
58. Stager, M. *et al.* (2021) Temperature heterogeneity correlates with intraspecific variation in physiological flexibility in a small endotherm. *Nat. Commun.* 12, 4401
59. Breuner, C.W. *et al.* (2008) In search of relationships between the acute adrenocortical response and fitness. *Gen. Comp. Endocrinol.* 157, 288–295
60. Schoenle, L.A. *et al.* (2021) Does variation in glucocorticoid concentrations predict fitness? A phylogenetic meta-analysis. *Gen. Comp. Endocrinol.* 300, 113611
61. Luttbeg, B. *et al.* (2021) Mathematical modeling reveals how the speed of endocrine regulation should affect baseline and stress-induced glucocorticoid levels. *Horm. Behav.* 136, 105059
62. Taborsky, B. *et al.* (2021) Towards an evolutionary theory of stress responses. *Trends Ecol. Evol.* 36, 39–48
63. Reed, J.M. *et al.* (2024) Is resilience a unifying concept for the biological sciences? *iScience* 27, 109478
64. May, R.M. (1973) *Stability and Complexity in Model Ecosystems*, Princeton University Press
65. Drury, C. (2020) Resilience in reef-building corals: the ecological and evolutionary importance of the host response to thermal stress. *Mol. Ecol.* 29, 448–465
66. Masel, J. and Trotter, M.V. (2010) Robustness and evolvability. *Trends Genet.* 26, 406–414
67. Romero, L.M. *et al.* (2015) Understanding stress in the healthy animal – potential paths for progress. *Stress* 18, 491–497
68. Guidi, J. *et al.* (2021) Allostatic load and its impact on health: a systematic review. *Psychother. Psychosom.* 90, 11–27
69. Eyck, H.J.F. *et al.* (2019) Effects of developmental stress on animal phenotype and performance: a quantitative review. *Biol. Rev.* 94, 1143–1160
70. Jiang, W.I. *et al.* (2024) Early-life stress triggers long-lasting organismal resilience and longevity via tetraspanin. *Sci. Adv.* 10, eadj3880
71. Nettle, D. and Bateson, M. (2015) Adaptive developmental plasticity: what is it, how can we recognize it and when can it evolve? *Proc. R. Soc. B Biol. Sci.* 282, 20151005
72. Costantini, D. *et al.* (2010) Ecological processes in a hormetic framework: hormesis in ecology. *Ecol. Lett.* 13, 1435–1447
73. Seebacher, F. *et al.* (2015) Physiological plasticity increases resilience of ectothermic animals to climate change. *Nat. Clim. Chang.* 5, 61–66
74. Trapp, S. *et al.* (2018) Stressful events as teaching signals for the brain. *Trends Cogn. Sci.* 22, 475–478

75. Hilker, M. *et al.* (2016) Priming and memory of stress responses in organisms lacking a nervous system. *Biol. Rev.* 91, 1118–1133
76. Taff, C.C. *et al.* (2024) Corticosterone exposure is associated with long-term changes in DNA methylation, physiology, and breeding decisions in a wild bird. *Mol. Ecol.* 33, e17456
77. Jakab, G. *et al.* (2005) Enhancing *Arabidopsis* salt and drought stress tolerance by chemical priming for its abscisic acid responses. *Plant Physiol.* 139, 267–274
78. De Kloet, E.R. and Joëls, M. (2024) The cortisol switch between vulnerability and resilience. *Mol. Psychiatry* 29, 20–34
79. Smeeth, D. *et al.* (2021) The role of epigenetics in psychological resilience. *Lancet Psychiatry* 8, 620–629